

Rack and Pinion Gripper

Design constraints:

- Cube of 100mm x 100mm x 100mm is being picked up
- Weight of gripper should be < 100g
- Weight of cube is about 1-2kg (2kg was taken into consideration)

Gripping force required:

- Payload mass = $m = 2\text{kg}$
- Gravity = $g = 9.81\text{m/s}^2$
- Weight = $W = mg = 19.62\text{N}$
- Safety Factor = $SF = 2$
- Coefficient of Friction = $f = 0.7$ (between rubber pads (gripper) and wooden box)

$$F(\text{frictional force required}) = W * SF = 39.24\text{N}$$

$$F(\text{clamping force required}) = (W * SF) / (2 * f) = 28.03\text{N}$$

$$F(\text{Each Finger}) = 14.015\text{N}$$

Torque Required:

Pinion Gear - 16 teeth, 1mm module, 20 degrees pressure angle, assumed/estimated mechanical efficiency of 0.7 for a 3D printed pinion gear

$$T(\text{required}) = F(\text{clamping force required}) * \text{Pinion Radius} / \text{Mechanical Efficiency} = 0.3203\text{Nm}$$

$$T(\text{required}) = 3.265\text{kg.cm}$$

EMAX ES3004 - Gives 3.5kg.cm torque while being just 20g in weight

Total Weight:

Assembly uses all materials in ABS but I wanted to use PETG but couldn't find it on the software so as per weight conversion, ABS has a density of 1.04g/cm^3 while PETG has a density of 1.22g/cm^3 . Hence the density ratio is 1.17.

$$\text{Assembly weight} = 62.691 + 2 * 1.841 = 66.373\text{g} \quad (2 \text{ rubber pads on the gripper surface})$$

$$\text{Actual weight} = 62.691 * 1.17 + 2 * 1.841 = 77.03\text{g}$$

$$\text{Final Weight with Motor} = 97.03\text{g}$$

